Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ann has 3 stickers. Ben has 4 times as many. How many does Ben have? Student's answer: 7

Solve the comparison problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 4 instead of multiplying. Times as many means 4 × 3 = 12.	**12 stickers**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rope is 5 meters long. Another rope is 2 times as long. How long is it? Student's answer: 7 meters

Solve the comparison problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 2 to 5. Two times as long means 2 × 5 = 10.	**10 meters**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Tom has 12 cards. Sam has 3 cards. How many times as many does Tom have? Student's answer: 9

Find how many times as many.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted. To find how many times as many, divide 12 by 3 = 4.	**4 times as many**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

book has 80 pages. Mia's has 20 pages. How many times as many pages does Ellen's have? Student's answ

Find how many times as many.

What did the student do wrong? Why do you think the student made this error?

The student subtracted the pages. Divide 80 by 20 to find the multiple: 4.

Solve the problem correctly. Show all of your work.

4 times as many

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dan is 6 years old. Lila is 3 times as old. How old is Lila? Student's answer: 9

Solve the comparison problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 3 to 6. Three times as old means 3 × 6 = 18.	**18 years old**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write '5 times as many as 4' as an equation. Student writes: 5 + 4

Write the equation.

What did the student do wrong? Why do you think the student made this error?

The student wrote addition. Times as many is multiplication: 5 × 4 = 20.

Solve the problem correctly. Show all of your work.

5 × 4 = 20

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ana has 8 books. Ben has 3 more than Ana. How many does Ben have? Student's answer: 24

Solve the comparison problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied instead of adding. 3 more than 8 is 8 + 3 = 11.	**11 books**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Ana has 8 books. Ben has 3 times as many. How many does Ben have? Student's answer: 11

Solve the comparison problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added instead of multiplying. 3 times as many is 3 × 8 = 24.	**24 books**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Tia has 15 shells. Max has 5 fewer. How many does Max have? Student's answer: 20

Solve the comparison problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added. Fewer means subtracting: 15 - 5 = 10.	**10 shells**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Lee has 3 books. Sara has 4 times as many. How many does Sara have? Student's answer: 7

Solve the comparison problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 4 to 3. Four times as many is 4 × 3 = 12.	**12 books**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pete ran 6 miles. Jo ran 4 more miles. How many did Jo run? Student's answer: 2

Solve the comparison problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted. Jo ran 4 more, so 6 + 4 = 10.	**10 miles**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Mo has 7 more than 2. Student writes: $7 \times 2 = 14$

Write and solve the equation.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used multiplication for a 'more than' comparison. 7 more than 2 is 2 + 7 = 9.	**9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A store has 4 boxes with 12 pencils each. 9 are sold. How many are left? Student's answer: 57

Solve the multi-step problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 9 to the total. 4 × 12 = 48, then 48 - 9 = 39.	**39 pencils**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Mia has 3 bags with 7 marbles each and 4 more marbles. How many in all? Student's answer: 84

Solve the multi-step problem.

What did the student do wrong? Why do you think the student made this error?

The student multiplied the extra marbles.
3 × 7 = 21, then 21 + 4 = 25.

Solve the problem correctly. Show all of your work.

25 marbles

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

re shared equally among 5 baskets. Each basket gets 2 more apples. How many in each basket? Student's

Solve the multi-step problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the 45 apples instead of dividing. 45 ÷ 5 = 9, then 9 + 2 = 11.	**11 apples**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Tickets cost $8 each. 6 tickets are bought with $60. What is the change? Student's answer: $108

Solve the multi-step problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the change by the cost. 6 × $8 = $48, then $60 - $48 = $12.	**$12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

4 classes have 23 students each, and 7 more students join. How many in all? Student's answer: 49

Solve the multi-step problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student left out the multiplication. 4 × 23 = 92, then 92 + 7 = 99.	**99 students**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve 3 × (5 + 4). Student's answer: 19

Solve using the order of operations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 3 to 5 + 4 without multiplying the whole group. Do the parentheses first: 3 × 9 = 27.	**27**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 15 a multiple of 3? Student's answer: no

Is it a multiple?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student checked the wrong way. 3 × 5 = 15, so 15 is a multiple of 3.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

List the factors of 12. Student's answer: 1, 2, 3, 4, 6

List all the factors.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student left out 12. Every number has itself as a factor, so 12 is also a factor.	**1, 2, 3, 4, 6, 12**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 7 prime? Student's answer: no

Prime or not?

What did the student do wrong? Why do you think the student made this error?

The student made a mistake. 7 has only two factors, 1 and 7, so it is prime.

Solve the problem correctly. Show all of your work.

Yes, 7 is prime.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 1 prime? Student's answer: yes

Prime or not?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A prime number has exactly two factors. The number 1 has only one factor, so it is not prime.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 36 prime or composite? Student's answer: prime

Prime or composite?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
36 has more than two factors, such as 2, 3, 4, 6, 9, and 12, so it is composite.	**Composite**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the even factors of 18. Student's answer: 2, 4, 6, 18

List the even factors.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student listed 4, which is not a factor of 18. The even factors are 2, 6, and 18.	**2, 6, 18**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pattern starts at 7 and adds 5: 7, 12, 17, 23. Find the error.

Check the pattern.

What did the student do wrong? Why do you think the student made this error?

The student wrote 23 instead of 22. The rule adds 5 each time, so 17 + 5 = 22.

Solve the problem correctly. Show all of your work.

22 should replace 23

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pattern starts at 2 and multiplies by 3: 2, 6, 18, 36. Find the error.

Check the pattern.

What did the student do wrong? Why do you think the student made this error?

The student multiplied 18 by 2 instead of 3. 18 × 3 = 54.

Solve the problem correctly. Show all of your work.

54 should replace 36

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says the rule for 3, 6, 9, 12 is 'add 4.'

Find the rule.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 4 instead of 3. The rule is add 3.	**Add 3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pattern: 5, 50, 500, 550. The rule is multiply by 10. Find the error.

Check the pattern.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 50 instead of multiplying. 500 × 10 = 5,000.	**5,000 should replace 550**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says the rule for 1, 3, 9, 27 is 'add 2.'

Find the rule.

What did the student do wrong? Why do you think the student made this error?

The student added 2, but the numbers do not go up by 2. The rule is multiply by 3.

Solve the problem correctly. Show all of your work.

Multiply by 3

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Pattern: 40, 36, 32, 28. Student says the next number is 22.

Find the next number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 6 instead of 4. The rule is subtract 4, so 28 - 4 = 24.	**24**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

In 500, the digit 5 is worth how much? Student's answer: 5

Find the value of the digit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the digit, not its place value. The 5 is in the hundreds place, so it is worth 500.	**500**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

10 times 300 = ? Student's answer: 30

Multiply by 10.

What did the student do wrong? Why do you think the student made this error?

The student did not shift the digits. Multiplying by 10 moves each digit one place to the left.

Solve the problem correctly. Show all of your work.

3,000

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many tens are in 4 hundreds? Student's answer: 4

Count the tens.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted the hundreds. Each hundred has 10 tens, so 4 hundreds is 40 tens.	**40 tens**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which digit is in the thousands place of 6,342? Student's answer: 3

Find the place value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student read the hundreds digit. The thousands digit in 6,342 is 6.	**6**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 70,000 in words. Student's answer: seven thousand

Write in words.

What did the student do wrong? Why do you think the student made this error?

The student left out the ten-thousands place. 70,000 is seventy thousand.

Solve the problem correctly. Show all of your work.

seventy thousand

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare the value of 7 in 7 tens and 7 in 7 hundreds. Student says 7 × 10 = 7.

Find the value.

What did the student do wrong? Why do you think the student made this error?

The student did not multiply by 10. 7 tens is 70, which is 10 times the value of 7 ones.

Solve the problem correctly. Show all of your work.

7 tens = 70

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

te 'four thousand twenty' as a numeral... Student's answer: 4,020 and says 'four thousand seventy-two' is 4,0

Write the number.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student mixed up the two numbers. Four thousand seventy-two is 4,072.	**4,072**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 'eighty-two thousand fifty-one' as a numeral. Student's answer: 82,501

Write the numeral.

What did the student do wrong? Why do you think the student made this error?

The student wrote the digits in the wrong places. Fifty-one is 051 at the end, so the number is 82,051.

Solve the problem correctly. Show all of your work.

82,051

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 4,523 and 4,532. Student writes: 4,523 > 4,532

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student reversed the symbol. Both have 4 thousands and 5 hundreds, but 2 tens is less than 3 tens.

Solve the problem correctly. Show all of your work.

4,523 < 4,532

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 9,105 and 9,015. Student writes: 9,105 < 9,015

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student reversed the symbol. 1 hundred is more than 0 hundreds, so 9,105 is greater.	**9,105 > 9,015**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 3,406 in expanded form. Student's answer: 3,000 + 40 + 6

Write in expanded form.

What did the student do wrong? Why do you think the student made this error?

The student wrote 40 instead of 400 for the hundreds place. 3,406 has 4 hundreds.

Solve the problem correctly. Show all of your work.

3,000 + 400 + 6

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 60,702 in expanded form. Student's answer: 6,000 + 700 + 2

Write in expanded form.

What did the student do wrong? Why do you think the student made this error?

The student left out the ten-thousands place. 60,702 has 6 ten-thousands, which is 60,000.

Solve the problem correctly. Show all of your work.

60,000 + 700 + 2

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 3,865 to the nearest hundred. Student's answer: 3,800

Round to the nearest hundred.

What did the student do wrong? Why do you think the student made this error?

The student rounded down. 3,865 is closer to 3,900.

Solve the problem correctly. Show all of your work.

3,900

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 47,650 to the nearest thousand. Student's answer: 47,000

Round to the nearest thousand.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student rounded down. 47,650 is halfway or more, so it rounds up to 48,000.	**48,000**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 638 to the nearest ten. Student's answer: 630

Round to the nearest ten.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student rounded down. 638 is closer to 640.	**640**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 2,500 to the nearest thousand. Student's answer: 2,000

Round to the nearest thousand.

What did the student do wrong? Why do you think the student made this error?

The student rounded down. Five hundred or more rounds up, so 2,500 rounds to 3,000.

Solve the problem correctly. Show all of your work.

3,000

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 9,949 to the nearest hundred. Student's answer: 9,000

Round to the nearest hundred.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student rounded to the thousand. 9,949 is closer to 9,900.	**9,900**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Round 125 to the nearest hundred. Student's answer: 200

Round to the nearest hundred.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student rounded up. 125 is closer to 100.	**100**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

4,326 + 2,518 = ? Student's answer: 6,734

Add.

What did the student do wrong? Why do you think the student made this error?

The student made an addition error in the tens or hundreds. 4,326 + 2,518 = 6,844.

Solve the problem correctly. Show all of your work.

6,844

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

7,000 - 3,456 = ? Student's answer: 4,456

Subtract.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a regrouping error. 7,000 - 3,456 = 3,544.	**3,544**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5,678 - 2,345 = ? Student's answer: 3,343

Subtract.

What did the student do wrong? Why do you think the student made this error?

The student made a subtraction error in the ones place. 5,678 - 2,345 = 3,333.

Solve the problem correctly. Show all of your work.

3,333

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

8,064 + 1,936 = ? Student's answer: 9,990

Add.

What did the student do wrong? Why do you think the student made this error?

The student made an addition error.
8,064 + 1,936 = 10,000.

Solve the problem correctly. Show all of your work.

10,000

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

45,612 - 9,384 = ? Student's answer: 36,338

Subtract.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a subtraction error in the tens place. 45,612 - 9,384 = 36,228.	**36,228**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

2,904 + 1,098 = ? Student's answer: 3,002

Add.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not add the thousands correctly. 2,904 + 1,098 = 4,002.	**4,002**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

6 × 2,000 = ? Student's answer: 8,000

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added instead of multiplying. 6 × 2 = 12, so 6 × 2,000 = 12,000.	**12,000**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

23 × 4 = ? Student's answer: 82

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a regrouping error. 4 × 3 = 12 and 4 × 20 = 80, so 23 × 4 = 92.	**92**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

17 × 25 = ? Student's answer: 325

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student left out a partial product. 17 × 25 = 425.	**425**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

38 × 12 = ? Student's answer: 336

Multiply.

What did the student do wrong? Why do you think the student made this error?

The student left out the 10 part of 12. 38 × 10 = 380, plus 38 × 2 = 76, so 456.

Solve the problem correctly. Show all of your work.

456

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3 × 1,234 = ? Student's answer: 3,612

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a regrouping error. 3 × 1,234 = 3,702.	**3,702**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

42 × 15 = ? Student's answer: 530

Multiply.

What did the student do wrong? Why do you think the student made this error?

The student left out a partial product. 42 × 10 = 420 and 42 × 5 = 210, so 630.

Solve the problem correctly. Show all of your work.

630

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

56 ÷ 4 = ? Student's answer: 13

Divide.

What did the student do wrong? Why do you think the student made this error?

The student made a division error. 4 × 14 = 56, so 56 ÷ 4 = 14.

Solve the problem correctly. Show all of your work.

14

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

75 ÷ 6 = ? Student's answer: 12 R 2

Divide with a remainder.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a remainder error. 6 × 12 = 72, and 75 - 72 = 3.	**12 R 3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

96 ÷ 8 = ? Student's answer: 11

Divide.

What did the student do wrong? Why do you think the student made this error?

The student made a division error. 8 × 12 = 96, so 96 ÷ 8 = 12.

Solve the problem correctly. Show all of your work.

12

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

84 ÷ 3 = ? Student's answer: 27

Divide.

What did the student do wrong? Why do you think the student made this error?

The student made a division error. 3 × 28 = 84, so 84 ÷ 3 = 28.

Solve the problem correctly. Show all of your work.

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Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

150 ÷ 5 = ? Student's answer: 3

Divide.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot the zero. 5 × 30 = 150, so 150 ÷ 5 = 30.	**30**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

65 ÷ 4 = ? Student's answer: 15 R 2

Divide with a remainder.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a division error. 4 × 16 = 64, so 65 ÷ 4 = 16 R 1.	**16 R 1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 2/3 equal to 4/6? Student's answer: no

Are the fractions equal?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not multiply both parts by 2. 2/3 = 4/6.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find a fraction equal to 3/4 with denominator 8. Student's answer: 3/8

Write an equivalent fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not change the numerator. Multiply both parts by 2 to get 6/8.	**6/8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 3/5 equal to 6/15? Student's answer: yes

Are the fractions equal?

What did the student do wrong? Why do you think the student made this error?

The student did not check the multiplication. 3 × 3 = 9, so 3/5 = 9/15, not 6/15.

Solve the problem correctly. Show all of your work.

No. 3/5 = 9/15.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 1/2 is equivalent to 2/3.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The fractions are not equal. 1/2 is less than 2/3.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find a fraction equal to 4/6 with denominator 3. Student's answer: 4/3

Write an equivalent fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student changed only the top number. 4/6 = 2/3, found by dividing both parts by 2.	**2/3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 5/10 = 1/5.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided only the bottom number. 5/10 = 1/2.	**No. 5/10 = 1/2.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 3/8 and 5/8. Student writes: 3/8 > 5/8

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student reversed the symbol. With the same denominator, 3/8 is less than 5/8.	**3/8 < 5/8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 2/3 and 3/4. Student writes: 2/3 > 3/4

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student reversed the comparison. 2/3 = 8/12 and 3/4 = 9/12, so 2/3 is less.

Solve the problem correctly. Show all of your work.

2/3 < 3/4

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 1/2 and 5/10. Student writes: 1/2 < 5/10

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student used the wrong symbol. 1/2 and 5/10 are equal fractions.

Solve the problem correctly. Show all of your work.

1/2 = 5/10

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 4/5 and 3/5. Student writes: 4/5 < 3/5

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student reversed the symbol. With the same denominator, 4/5 is greater.

Solve the problem correctly. Show all of your work.

4/5 > 3/5

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 7/8 and 1. Student writes: 7/8 > 1

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student reversed the symbol. 7/8 is less than 1 whole.	**7/8 < 1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 2/6 and 1/3. Student writes: 2/6 < 1/3

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student thought the fractions were different. 2/6 = 1/3, so they are equal.	**2/6 = 1/3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 3/5 as a sum of unit fractions. Student's answer: 1/5 + 2/5

Write as a sum of unit fractions.

What did the student do wrong? Why do you think the student made this error?

The student used a non-unit fraction. A unit fraction has a numerator of 1, so 3/5 = 1/5 + 1/5 + 1/5.

Solve the problem correctly. Show all of your work.

1/5 + 1/5 + 1/5

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3/4 + 1/4 = ? Student's answer: 4/8

Add the fractions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the denominators. When denominators match, keep the denominator: 4/4 = 1.	**1**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5/6 - 2/6 = ? Student's answer: 3/0

Subtract the fractions.

What did the student do wrong? Why do you think the student made this error?

The student subtracted the denominators. Keep the denominator: 5/6 - 2/6 = 3/6, which simplifies to 1/2.

Solve the problem correctly. Show all of your work.

3/6 = 1/2

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 2 1/3 as an improper fraction. Student's answer: 2/3

Write as an improper fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student kept only the fraction part. 2 1/3 = 7/3.	**7/3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

7/8 - 5/8 = ? Student's answer: 12/8

Subtract the fractions.

What did the student do wrong? Why do you think the student made this error?

The student added the numerators. 7 - 5 = 2, so 7/8 - 5/8 = 2/8 = 1/4.

Solve the problem correctly. Show all of your work.

2/8 = 1/4

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

2/5 + 2/5 = ? Student's answer: 4/10

Add the fractions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the denominators. Keep the denominator: 2/5 + 2/5 = 4/5.	**4/5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$3 \times 1/4 = ?$ Student's answer: 3/12

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the denominator too. Multiply only the numerator: 3 × 1 = 3, so the answer is 3/4.	**3/4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$4 \times 2/5 = ?$ Student's answer: 6/5

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 2 instead of multiplying. 4 × 2 = 8, so the answer is 8/5.	**8/5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5 × 1/8 = ? Student's answer: 5/40

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the denominator. Keep the denominator: 5/8.	**5/8**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$2 \times 3/6 = ?$ Student's answer: 6/12

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the denominator. $2 \times 3 = 6$, so $6/6 = 1$.	**1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A recipe uses 1/3 cup of milk, 3 times. How much milk? Student's answer: 3/9 cup

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the denominator. **3 × 1/3 = 3/3 = 1 cup.**	**1 cup**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$2/3 \times 3 = ?$ Student's answer: $6/9$

Multiply.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the denominator. $2 \times 3 = 6$, and $6/3 = 2$.	**2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 3/10 as hundredths. Student's answer: 3/100

Write with denominator 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student changed the denominator without the numerator. Multiply both by 10: 30/100.	**30/100**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 7/10 as hundredths. Student's answer: 70/10

Write with denominator 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student kept the denominator at 10. To get hundredths, multiply both by 10: 70/100.	**70/100**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.4 as a fraction. Student's answer: 4/100

Write as a fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the wrong place value. The 4 is in the tenths place, so 0.4 = 4/10.	**4/10**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.09 as a fraction. Student's answer: 9/10

Write as a fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the wrong place value. The 9 is in the hundredths place, so 0.09 = 9/100.	**9/100**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 6/100 as a decimal. Student's answer: 0.6

Write as a decimal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student moved the digit to the wrong place. 6/100 is 0.06.	**0.06**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Add 3/10 + 45/100. Student's answer: 48/110

Add the fractions.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the denominators. Write 3/10 as 30/100 first, then add: 30/100 + 45/100 = 75/100.	**75/100**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.7 as a fraction. Student's answer: 7/100

Write as a fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used hundredths. The 7 is in the tenths place, so 0.7 = 7/10.	**7/10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 4/10 as a decimal. Student's answer: 0.04

Write as a decimal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the wrong place value. 4 tenths is 0.4.	**0.4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.52 in expanded form. Student's answer: 5 + 2

Write in expanded form.

What did the student do wrong? Why do you think the student made this error?

The student did not show the place values. 0.52 = 5/10 + 2/100.

Solve the problem correctly. Show all of your work.

5/10 + 2/100

100

Find the Error

4.NF.6

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 0.3 and 0.30. Student writes: 0.3 > 0.30

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student thought the extra zero makes a difference. 0.3 and 0.30 are equal.

Solve the problem correctly. Show all of your work.

0.3 = 0.30

Original task. Format inspired by error analysis routines.

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Find the Error

4.NF.6

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 23/100 as a decimal. Student's answer: 2.3

Write as a decimal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student put the decimal in the wrong place. 23/100 = 0.23.	**0.23**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.6 as hundredths. Student's answer: 6/100

Write with denominator 100.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student kept the value of the tenths place as hundredths. 0.6 = 60/100.	**60/100**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 0.4 and 0.37. Student writes: 0.4 < 0.37

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student compared the digits from the wrong side. 0.4 is 0.40, which is greater than 0.37.

Solve the problem correctly. Show all of your work.

0.4 > 0.37

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 0.6 and 0.60. Student writes: 0.6 > 0.60

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student thought the extra zero made a difference. 0.6 and 0.60 are equal.

Solve the problem correctly. Show all of your work.

0.6 = 0.60

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 0.25 and 0.5. Student writes: 0.25 > 0.5

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student compared 25 and 5 as whole numbers. 0.5 is 0.50, which is greater than 0.25.	**0.25 < 0.5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 0.8 and 0.09. Student writes: 0.8 < 0.09

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student reversed the symbol. 0.8 has 8 tenths, which is more than 9 hundredths.

Solve the problem correctly. Show all of your work.

0.8 > 0.09

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Order from least to greatest: 0.5, 0.45, 0.6. Student's answer: 0.6, 0.5, 0.45

Order the decimals.

What did the student do wrong? Why do you think the student made this error?

The student ordered them from greatest to least. The order from least to greatest is 0.45, 0.5, 0.6.

Solve the problem correctly. Show all of your work.

0.45, 0.5, 0.6

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 0.7 and 7/10. Student writes: 0.7 > 7/10

Use >, <, or =

What did the student do wrong? Why do you think the student made this error?

The student thought the forms were different. 0.7 and 7/10 are the same amount.

Solve the problem correctly. Show all of your work.

0.7 = 7/10

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3 feet = ? inches. Student's answer: 3 inches

Convert the units.

What did the student do wrong? Why do you think the student made this error?

The student kept the same number. There are 12 inches in 1 foot, so 3 feet = 36 inches.

Solve the problem correctly. Show all of your work.

36 inches

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

2 meters = ? centimeters. Student's answer: 20 cm

Convert the units.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 10 instead of 100. There are 100 cm in 1 meter, so 2 meters = 200 cm.	**200 cm**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5 kg = ? grams. Student's answer: 500 g

Convert the units.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 100 instead of 1,000. There are 1,000 g in 1 kg, so 5 kg = 5,000 g.	**5,000 g**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

4 hours = ? minutes. Student's answer: 40 minutes

Convert the units.

What did the student do wrong? Why do you think the student made this error?

The student used 10 instead of 60. There are 60 minutes in 1 hour, so 4 hours = 240 minutes.

Solve the problem correctly. Show all of your work.

240 minutes

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

1 yard = ? feet. Student's answer: 2 feet

Convert the units.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the wrong number. There are 3 feet in 1 yard.	**3 feet**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

6 liters = ? milliliters. Student's answer: 600 mL

Convert the units.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used 100 instead of 1,000. There are 1,000 mL in 1 liter, so 6 liters = 6,000 mL.	**6,000 mL**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A ribbon is 6 feet long. Ben uses 18 inches. How many inches are left? Student's answer: 18 inches

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave the amount used. 6 feet is 72 inches, and 72 - 18 = 54 inches left.	**54 inches**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A bottle holds 2 L 300 mL. It gives away 500 mL. How much is left? Student's answer: 2 L 200 mL

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted the liters only. 2,300 mL - 500 mL = 1,800 mL, or 1 L 800 mL.	**1 L 800 mL**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

$3.45 + $2.80 = ? Student's answer: $5.25

Add the money.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made an addition error. $3.45 + $2.80 = $6.25.	**$6.25**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

3 hours 40 minutes + 1 hour 35 minutes = ? Student's answer: 4 h 75 min

Add the time.

What did the student do wrong? Why do you think the student made this error?

The student did not regroup 60 minutes into an hour. 75 minutes is 1 h 15 min, so the total is 5 h 15 min.

Solve the problem correctly. Show all of your work.

5 hours 15 minutes

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sack weighs 5 kg 200 g. 1 kg 500 g is removed. What is left? Student's answer: 4 kg 300 g

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made a subtraction error. 5,200 g - 1,500 g = 3,700 g, which is 3 kg 700 g.	**3 kg 700 g**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 4 L pitcher has 1,500 mL poured out. How many mL are left? Student's answer: 5,500 mL

Solve the word problem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the amount poured out. 4,000 mL − 1,500 mL = 2,500 mL.	**2,500 mL**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a rectangle 9 by 4. Student's answer: 26 square units

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student found the perimeter. Area is 9 × 4 = 36 square units.

Solve the problem correctly. Show all of your work.

36 square units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the perimeter of a rectangle 7 by 3. Student's answer: 21

Find the perimeter.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the sides. Perimeter is 7 + 3 + 7 + 3 = 20.	**20 units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the perimeter of a square with side 5. Student's answer: 10

Find the perimeter.

What did the student do wrong? Why do you think the student made this error?

The student added only two sides.
Perimeter is 5 + 5 + 5 + 5 = 20.

Solve the problem correctly. Show all of your work.

20 units

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle has area 48 and width 6. Find the length. Student's answer: 42

Find the missing length.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted. Length = area ÷ width = 48 ÷ 6 = 8.	**8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangle has perimeter 24 and length 8. Find the width. Student's answer: 16

Find the missing width.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the wrong number. 2 × (8 + width) = 24, so width = 4.	**4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the area of a 12 by 5 rectangle. Student's answer: 17

Find the area.

What did the student do wrong? Why do you think the student made this error?

The student added the sides. Area is 12 × 5 = 60.

Solve the problem correctly. Show all of your work.

60 square units

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line plot has 3 X's at 1/2 inch and 2 X's at 3/4 inch. Total length? Student's answer: 2 1/4 inches

Find the total length.

What did the student do wrong? Why do you think the student made this error?

The student added the values wrong. 3 × 1/2 = 1 1/2, and 2 × 3/4 = 1 1/2, so the total is 3 inches.

Solve the problem correctly. Show all of your work.

3 inches

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

4 X's are each 1/8 inch. Total length? Student's answer: 4/64 inch

Find the total length.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the denominators. Keep the denominator: 4 × 1/8 = 4/8 = 1/2 inch.	**1/2 inch**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line plot has 2 X's at 1/4 inch and 1 X at 3/8 inch. Total length? Student's answer: 3/12 inch

Find the total length.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the denominators. 2/4 + 3/8 = 4/8 + 3/8 = 7/8 inch.	**7/8 inch**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the difference between 5/8 and 1/8. Student's answer: 6/8

Find the difference.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added instead of subtracting. 5/8 - 1/8 = 4/8 = 1/2.	**4/8 = 1/2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is longer, 5/8 inch or 3/4 inch? Student's answer: 5/8 inch

Compare the lengths.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not compare with the same denominator. 3/4 = 6/8, which is longer than 5/8.	**3/4 inch**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 1/2 inch is shorter than 1/4 inch.

Is this right?

What did the student do wrong? Why do you think the student made this error?

The student reversed the comparison. 1/4 is shorter than 1/2.

Solve the problem correctly. Show all of your work.

No. 1/4 inch is shorter.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 100° angle is acute or obtuse? Student's answer: acute

Acute or obtuse?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student confused the terms. Acute angles are less than 90°, and 100° is more than 90°, so it is obtuse.	**Obtuse**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right angle measures how many degrees? Student's answer: 180°

Find the measure.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave a straight angle. A right angle measures 90°.	**90°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 45° angle is a right angle. Is this right?

Is this right?

What did the student do wrong? Why do you think the student made this error?

The student confused a right angle with an acute angle. 45° is less than 90°, so it is acute.

Solve the problem correctly. Show all of your work.

No. 45° is an acute angle.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A full circle measures how many degrees? Student's answer: 180°

Find the measure.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student gave a straight angle. A full circle is 360°.	**360°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says an angle that measures 180° is a right angle.

Is this right?

What did the student do wrong? Why do you think the student made this error?

The student confused a straight angle with a right angle. 180° is a straight angle.

Solve the problem correctly. Show all of your work.

No. 180° is a straight angle.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 30° angle is obtuse. Is this right?

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student confused the terms. 30° is less than 90°, so it is acute.	**No. 30° is acute.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

gle's ray lines up with the 0 on the inner scale and the other ray points to 50 on the inner scale. Student rea

Read the angle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student read the outer scale. Use the scale that starts at 0 on the same ray, which gives 50°.	**50°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student draws a 90° angle but says it measures 45°.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student misread the protractor. A 90° angle is a right angle, and it measures 90°.	**No. The angle measures 90°.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student measures an angle and reads 75°, then says it is 105° because they read the other scale.

Read the angle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student read the wrong scale. Use the scale that starts at 0 on the ray that lines up with the protractor's base.	**75°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a 120° angle measures 60°.

Is this right?

What did the student do wrong? Why do you think the student made this error?

The student read the protractor incorrectly. A 120° angle measures 120°.

Solve the problem correctly. Show all of your work.

No. The angle is 120°.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a 35° angle is 145° because they read the wrong scale.

Read the angle.

What did the student do wrong? Why do you think the student made this error?

The student used the wrong scale. Read from the 0 on the same ray, which gives 35°.

Solve the problem correctly. Show all of your work.

35°

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says 40° + 25° = 55° when they add the angles.

Add the angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made an addition error. 40° + 25° = 65°.	**65°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Angle parts are 30° and 45°. What is the total? Student's answer: 65°

Add the angle parts.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made an addition error. 30° + 45° = 75°.	**75°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles together make 90°, and one is 35°. What is the other? Student's answer: 65°

Find the missing angle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added instead of subtracting. 90° - 35° = 55°.	**55°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A 120° angle is split into 50° and 60°. Student says this is correct.

Is this right?

What did the student do wrong? Why do you think the student made this error?

The parts add to 110°, not 120°. A 10° part is missing.

Solve the problem correctly. Show all of your work.

No. The parts add to 110°, so 10° is missing.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A straight line is 180°. One angle is 75°. What is the other angle? Student's answer: 75°

Find the missing angle.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student copied the given angle. 180° - 75° = 105°.	**105°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two angles measure 20° and 40°. Student says the angle is 70°.

Find the total.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student made an addition error. 20° + 40° = 60°.	**60°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a 100° angle is made of two 45° angles.

Is this right?

What did the student do wrong? Why do you think the student made this error?

Two 45° angles make 90°, not 100°.

Solve the problem correctly. Show all of your work.

No. Two 45° angles make 90°.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a ray has two endpoints. Is this right?

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A ray starts at one endpoint and goes on forever in one direction.	**No. A ray has one endpoint.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says parallel lines meet at one point.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Parallel lines never meet. They stay the same distance apart.	**No. Parallel lines never meet.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says perpendicular lines form a 45° angle.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Perpendicular lines form right angles, which measure 90°.	**No. Perpendicular lines form 90° angles.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a line segment goes on forever.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A line segment has two endpoints and a fixed length. A line goes on forever in both directions.	**No. A line segment has two endpoints.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a line has one endpoint.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A line has no endpoints. It goes on forever in both directions.	**No. A line has no endpoints.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a right angle is a 180° angle.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A right angle is 90°. A 180° angle is a straight angle.	**No. A right angle is 90°.**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a triangle with one right angle is an equilateral triangle.

Name the triangle.

What did the student do wrong? Why do you think the student made this error?

An equilateral triangle has three equal sides and no right angle. A triangle with one right angle is a right triangle.

Solve the problem correctly. Show all of your work.

A right triangle

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says an acute triangle has one obtuse angle.

Is this right?

What did the student do wrong? Why do you think the student made this error?

An acute triangle has all three angles acute. A triangle with one obtuse angle is an obtuse triangle.

Solve the problem correctly. Show all of your work.

No. All angles in an acute triangle are acute.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a rectangle has 4 equal sides.

Is this right?

What did the student do wrong? Why do you think the student made this error?

A rectangle has 2 pairs of equal sides. Only a square has 4 equal sides.

Solve the problem correctly. Show all of your work.

No. A rectangle has 2 pairs of equal sides.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a right triangle has 2 right angles.

Is this right?

What did the student do wrong? Why do you think the student made this error?

A triangle can have only one right angle, since the other two angles must add to 90°.

Solve the problem correctly. Show all of your work.

No. A right triangle has one right angle.

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says an obtuse triangle has 3 acute angles.

Is this right?

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
An obtuse triangle has one obtuse angle and two acute angles.	**No. It has 2 acute angles and 1 obtuse angle.**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Student says a rhombus always has right angles.

Is this right?

What did the student do wrong? Why do you think the student made this error?

A rhombus has 4 equal sides, but its angles do not have to be right angles. Only a square has right angles.

Solve the problem correctly. Show all of your work.

No. A rhombus does not always have right angles.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many lines of symmetry does a square have? Student's answer: 2

Count the lines of symmetry.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only the vertical and horizontal lines. A square also has 2 diagonal lines, for 4 in all.	**4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many lines of symmetry does a rectangle that is not a square have? Student's answer: 4

Count the lines of symmetry.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted as if it were a square. A non-square rectangle has 2 lines of symmetry.	**2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many lines of symmetry does an equilateral triangle have? Student's answer: 1

Count the lines of symmetry.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student missed two lines. An equilateral triangle has 3 lines of symmetry.	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many lines of symmetry does a circle have? Student's answer: 0

Count the lines of symmetry.

What did the student do wrong? Why do you think the student made this error?

The student thought a circle has no symmetry. A circle has infinitely many lines of symmetry.

Solve the problem correctly. Show all of your work.

Infinitely many

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

w many lines of symmetry does a parallelogram that is not a rhombus or rectangle have? Student's answe

Count the lines of symmetry.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted the lines of a rhombus or rectangle. A general parallelogram has no lines of symmetry.	**0**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

How many lines of symmetry does a regular hexagon have? Student's answer: 3

Count the lines of symmetry.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student counted only half the lines. A regular hexagon has 6 lines of symmetry.	**6**

Original task. Format inspired by error analysis routines.